CAN Bus Logic & Arbitration

Wired-AND Logic

Sender 1	0	1	0	1	0	1	0	1
Sender 2	0	0	1	1	0	0	1	1
Sender 3	0	0	0	0	1	1	1	1
CAN Bus	0	0	0	0	0	0	0	1

Arbitration Logic

CAN Node/ Send Level	CAN Bus Level	Reaction of CAN Node
0	0	Routing
0	1	Error: Stop
1	0	Stop: Receive
1	1	Routing

1"

R

CAN Bus

CAN Node 1

CAN Node 2

CAN Node 3

S_1 R_1

0 0

S_2 R_2

1 0

S_3 R_3

1 0

Reaction

Reaction

Reaction

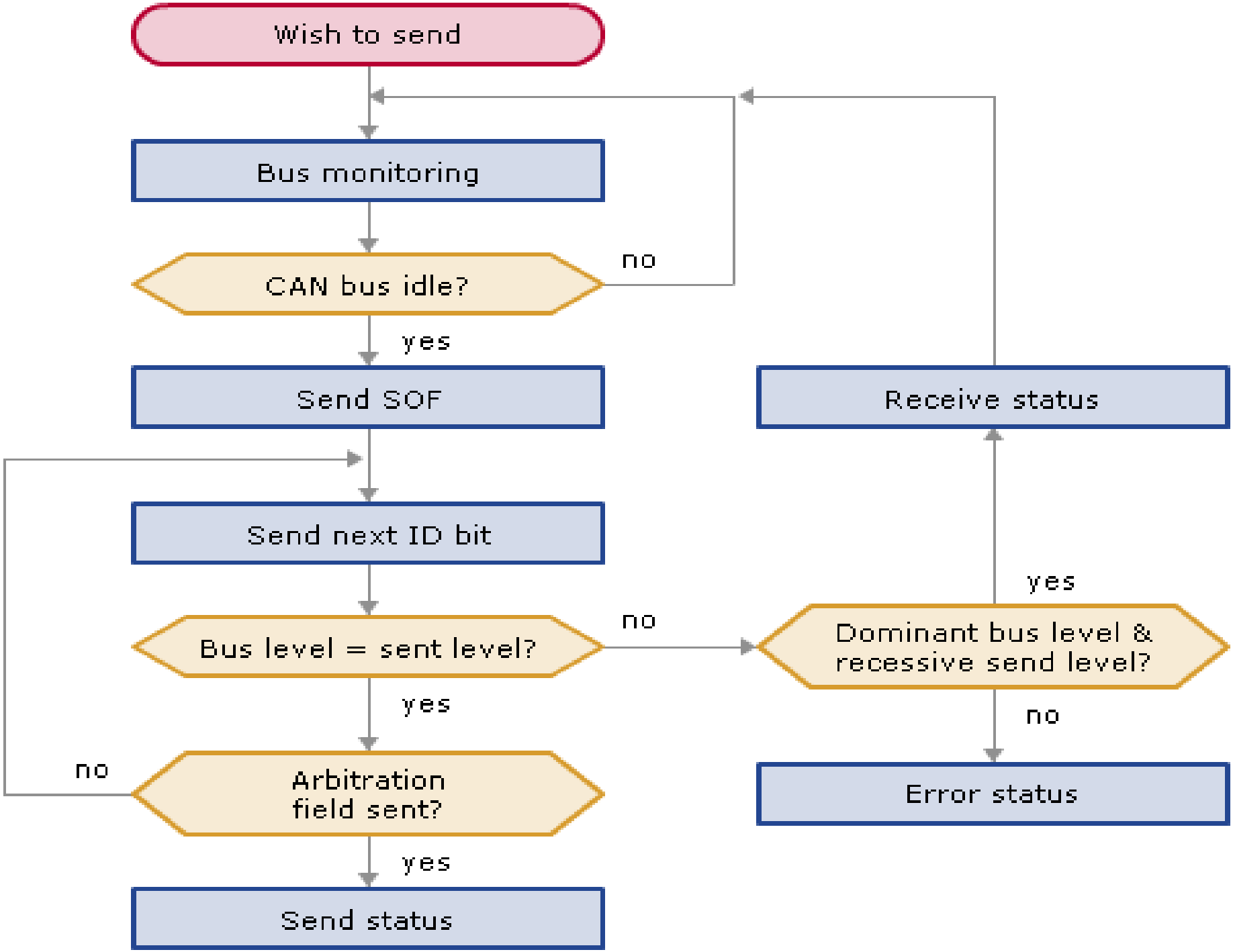
Routing

Stop: Receive

Stop: Receive

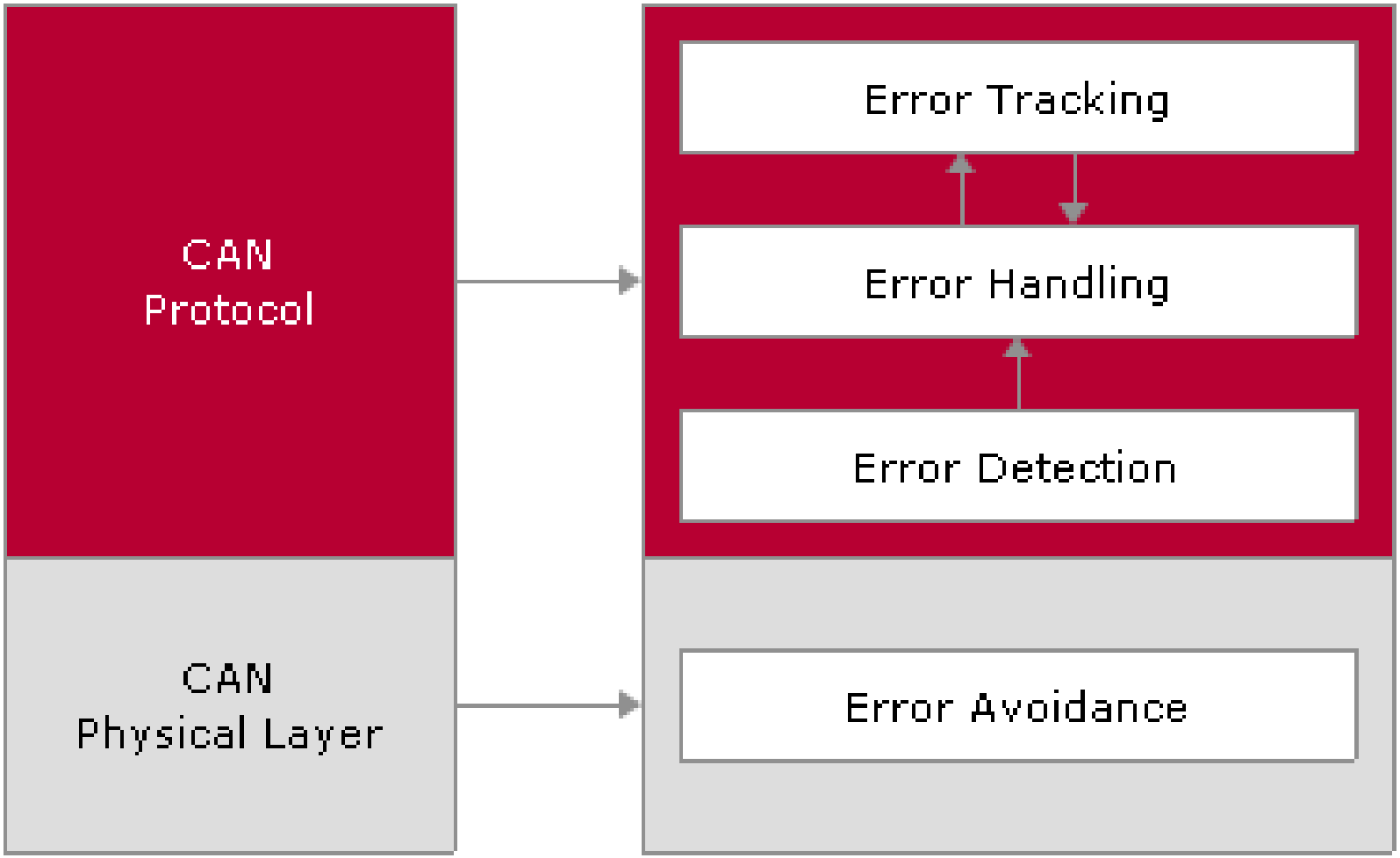
CAN Bus Access

PROCEDURE : CSMA/CD +AMP


Wish to send
Bus monitoring
CAN bus idle?
no
yes
Send SOF
Send next ID bit
Bus level = sent level?
no
yes
Arbitration field sent?
no
yes
Send status
Receive status
Dominant bus level & recessive send level?
yes
no
Error status

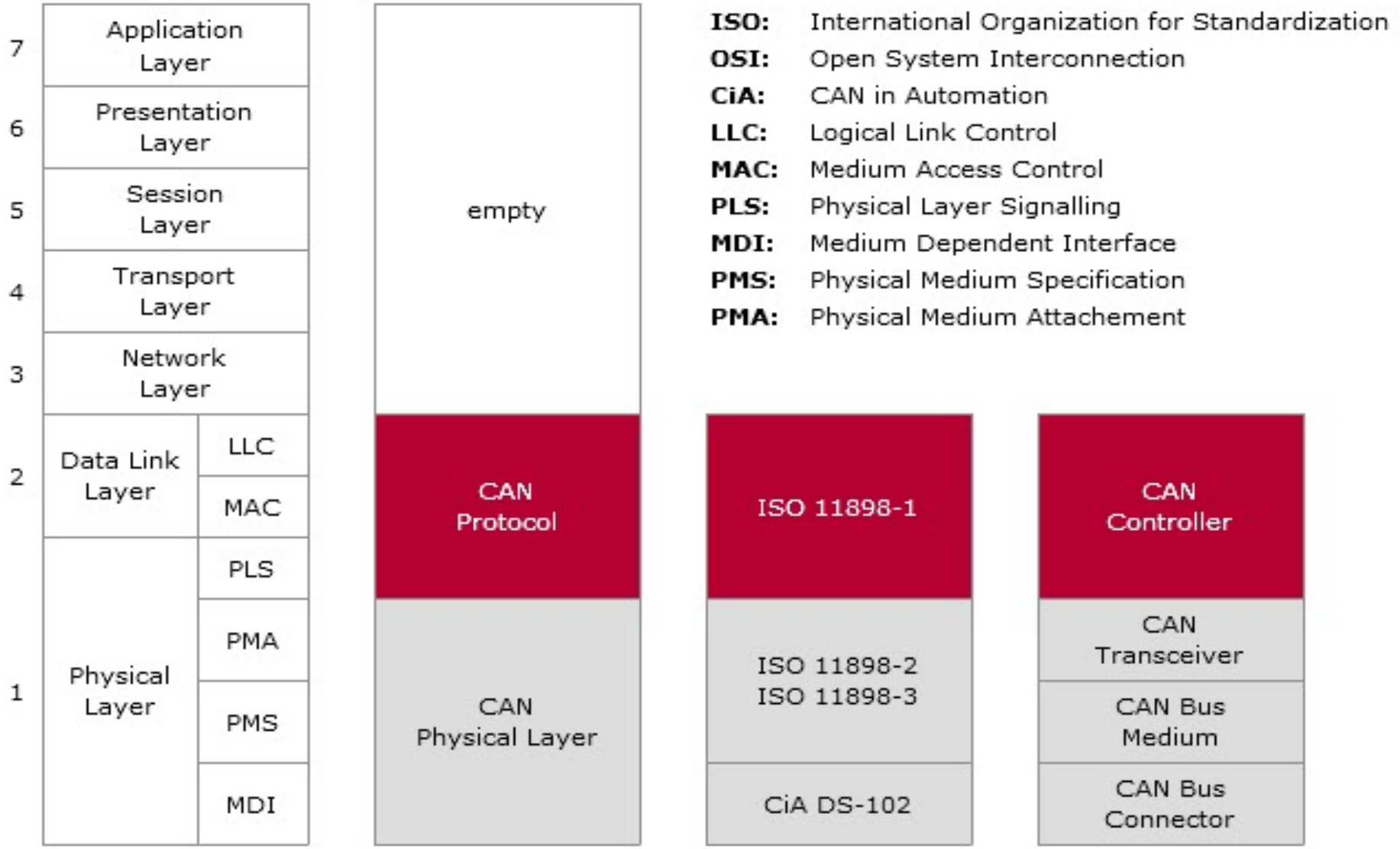
Data Protection in the CAN Network

Principle of Data Protection

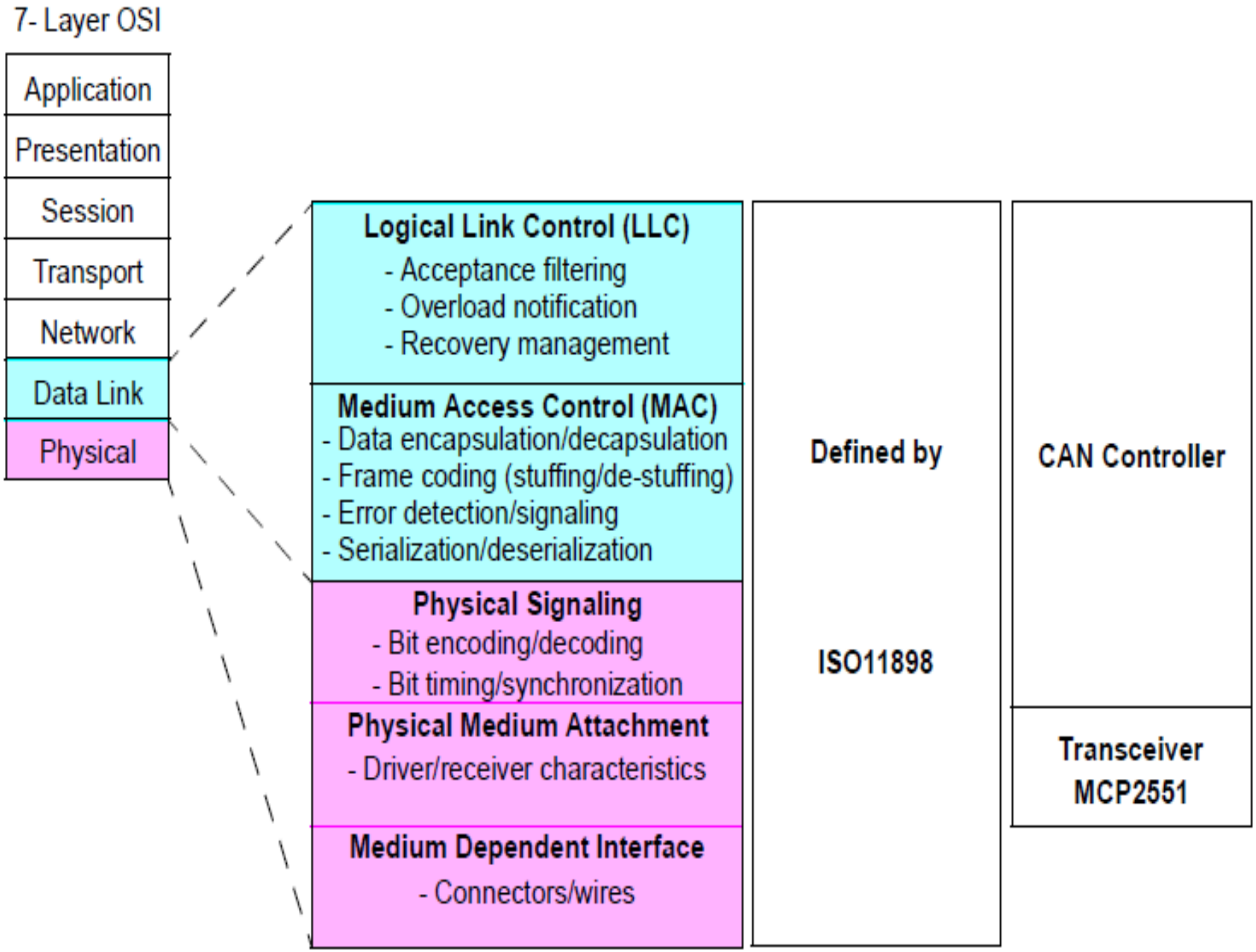


CAN Introduction

Standard and Implementation



ISO: International Organization for Standardization
OSI: Open System Interconnection
CiA: CAN in Automation
LLC: Logical Link Control
MAC: Medium Access Control
PLS: Physical Layer Signalling
MDI: Medium Dependent Interface
PMS: Physical Medium Specification
PMA: Physical Medium Attachement


7- Layer OSI
Application
Presentation
Session
Transport
Network
Data Link
Physical
Logical Link Control (LLC)
- Acceptance filtering
- Overload notification
- Recovery management
Medium Access Control (MAC)
- Data encapsulation/decapsulation
- Frame coding (stuffing/de-stuffing)
- Error detection/signaling
- Serialization/deserialization
Physical Signaling
- Bit encoding/decoding
- Bit timing/synchronization
Physical Medium Attachment
- Driver/receiver characteristics
Medium Dependent Interface
- Connectors/wires
Defined by
ISO11898
CAN Controller
Transceiver
MCP2551